RECLAIM — Continuous-Sweep Pump Sizing (Peak Non-Con

Worst-case procurement envelope. Yellow cells are inputs — change them and recalc. replaces every assumed input.

INPUTS (set the levers — conservative end shown)

Feedstock mass, m_feed	750	g
Gas yield fraction, y_gas	0.65	—
Active evolution window, t_active	15	min
Peak-to-average factor, f	3	—
Mean molar mass, M_bar	20	g/mol
Pump inlet temp, T_in	290	K
Pump inlet pressure, P_in	0.75	atm
Carrier / inert sweep, Q_carrier	0.2	L/min
Design margin	1.5	×
Gas constant, R′	0.08206	L·atm/mol·K

CALCULATION (live formulas)

1. Gas mass, m_gas	487.50	g
2. Avg gas rate, m_dot_avg	32.50	g/min
3. Peak gas rate, m_dot_peak	97.50	g/min
4. Peak molar rate, n_dot_peak	4.88	mol/min
5. Molar volume at inlet, V_m	31.73	L/mol
6. Peak non-cond. flow, Q_noncond	**154.68**	L/min
7. Required (with carrier+margin)	**232.32**	L/min

PUMP COMPARISON (KNF LABOPORT family, verified specs)

Model	Max flow	Derated ~70
UN 810.3 FTP (current BOM)	10	7.0
UN 820.3 FTP	20	14.0
UN 840.3 FTP	34	23.8

MAX BATCH BACK-SOLVER (how big a batch can a given pump k

Target throughput (derated pump)	7	L/min
Max feedstock at this throughput	**34**	g

SCENARIO REFERENCE (peak Q_noncond at 750 g, for context)

Worst-case (y=.65,t=15,f=3,M=20)	≈ 124 L/min
Central (y=.55,t=30,f=2.5,M=30)	≈ 29 L/min
Optimistic (y=.50,t=45,f=2,M=40)	≈ 11 L/min

Even the optimistic peak (~11 L/min) exceeds the 810.3 headline. The 750 g full ba batch.

densable Flow)

Output is a CEILING, not a prediction; shakedown exhaust-meter run

Batch charge of LDPE
Fraction of feed to non-condensable gas (0.50–0.65)
BIGGEST lever. Short end = conservative (15–40 min)
Single-peak DTG shape (3–4)
Low = H2-rich = worst case (15–35)
POST-condenser, ~300–310 K — NOT reactor T
Lowest setpoint = worst case
Run-phase sweep; tiny vs gas peak (0.04–0.2)
Absorbs M_bar error, Boudouard, transients (1.5–2)
Constant

m_feed × y_gas
m_gas / t_active
f × m_dot_avg
m_dot_peak / M_bar
R′ × T_in / P_in
n_dot_peak × V_m — what the pump head sees
margin × (Q_noncond + Q_carrier)

Verdict vs peak Q_noncond (B21)
FAIL — short by 22.1× of derated capacity
FAIL — short by 11.0× of derated capacity
FAIL — short by 6.5× of derated capacity

eep up with?)

e.g. 810.3 derated ≈ 7 L/min. Solve for max m_feed at peak.

Keep shakedown batch at/under this. ≤50 g is safe across all assumptions.

atch needs a pump well beyond this family — or a much smaller